World energy supplies

Web links

- https://en.wikipedia.org/wiki/Energy_policy_of_India
- https://energypedia.info/wiki/India_Energy_Situation
- https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.939.9423&rep=rep1&type=pdf
- https://iea.blob.core.windows.net/assets/fd69e584-f43f-400b-9702-f5a6dc9c3156/WEO2020-Launch-Presentation.pdf
- https://www.euro-fusion.org/fileadmin/user_upload/Archive/wp-content/uploads/2015/02/ETM_WP12_GS_FINAL.pdf
- https://en.wikipedia.org/wiki/Electricity_sector_in_India
- https://cea.nic.in/wp-content/uploads/pdm/2020/12/growth_2020.pdf

TOTAL INSTALLED POWER GENERATION CAPACITY (END OF APRIL 2017)

SOURCE	UTILITIES CAPACITY (MW)	%
Coal	194,402.88	59.9
Hydroelectricity	44,594.42	14.0
Renewable energy source	50,018.00	15.9
Natural Gas	25,329.38	8.1
Nuclear	6,780.00	1.8
Oil	837.63	0.3
Total	**329,204.53**	

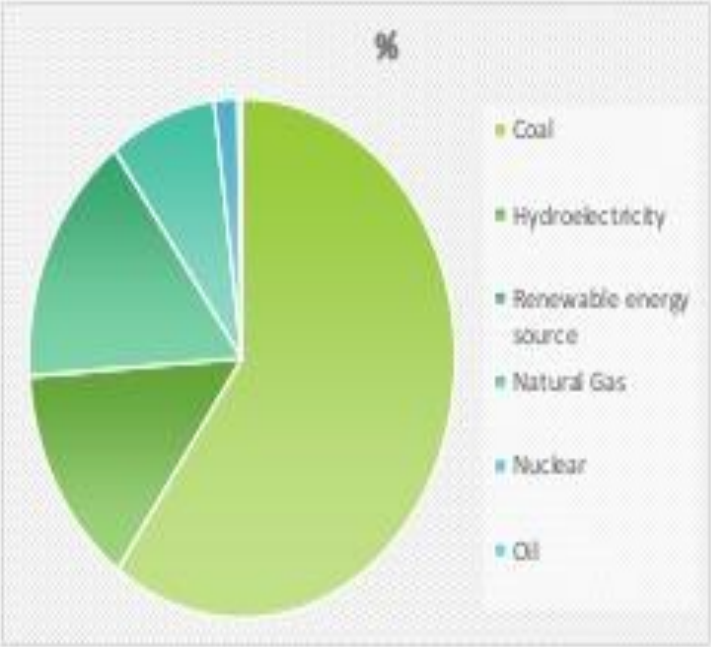
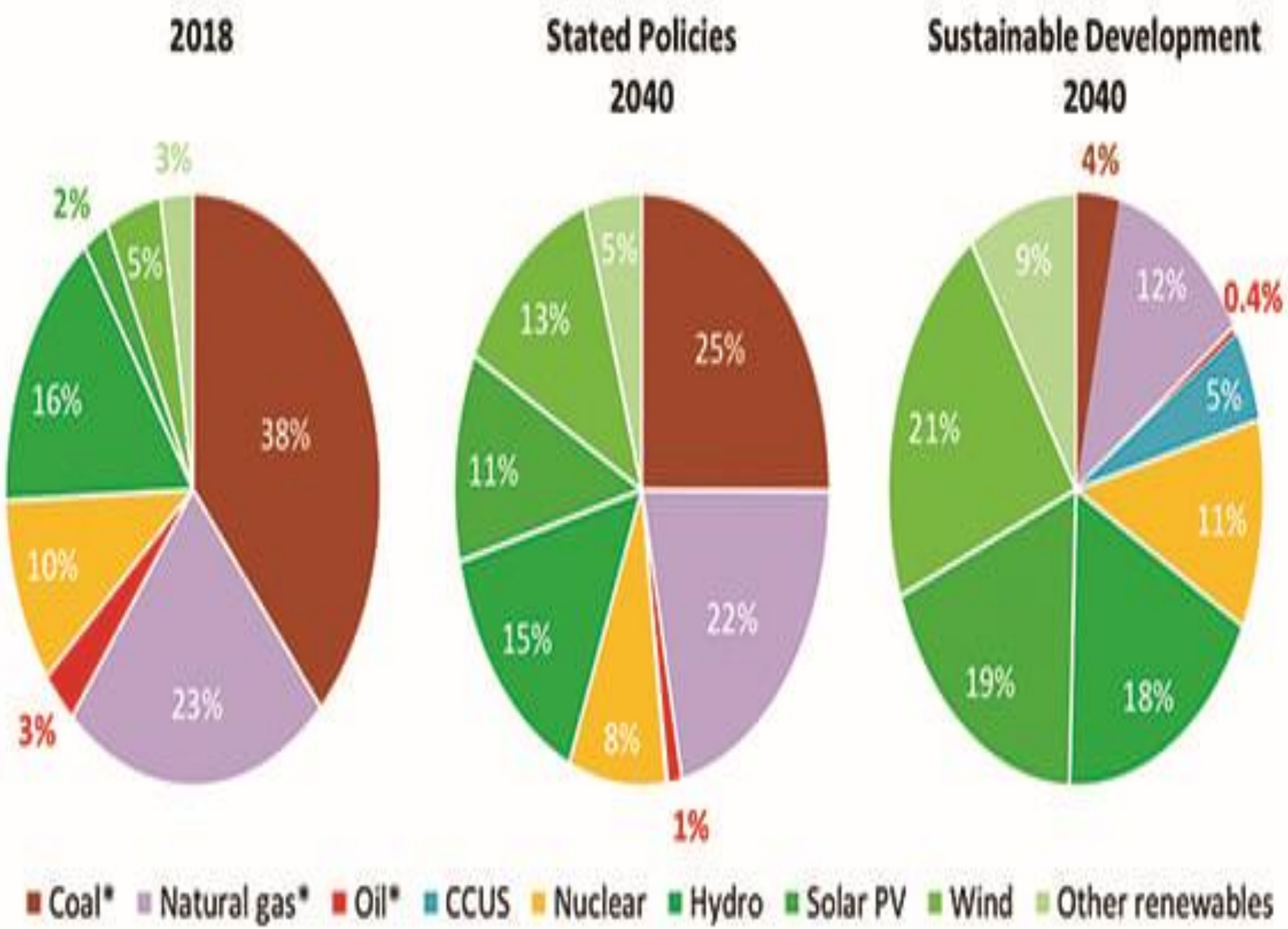
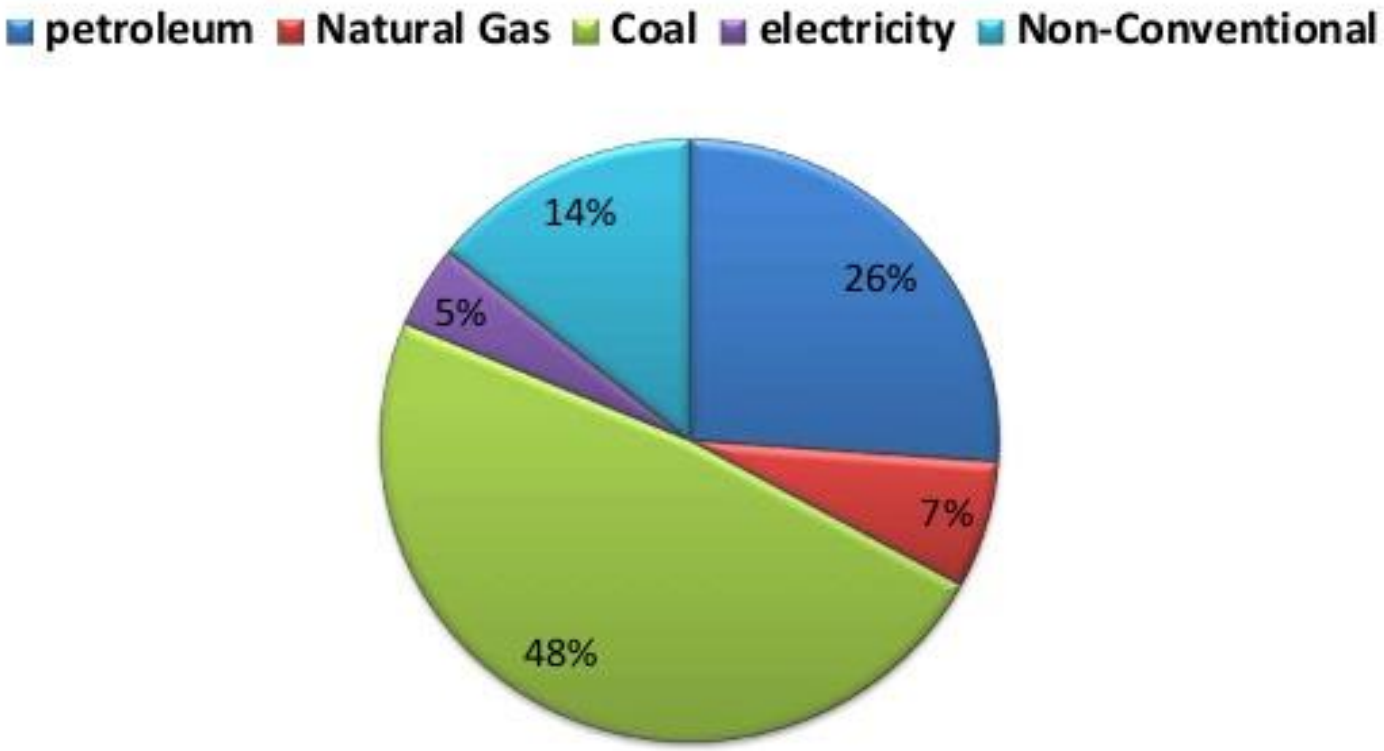
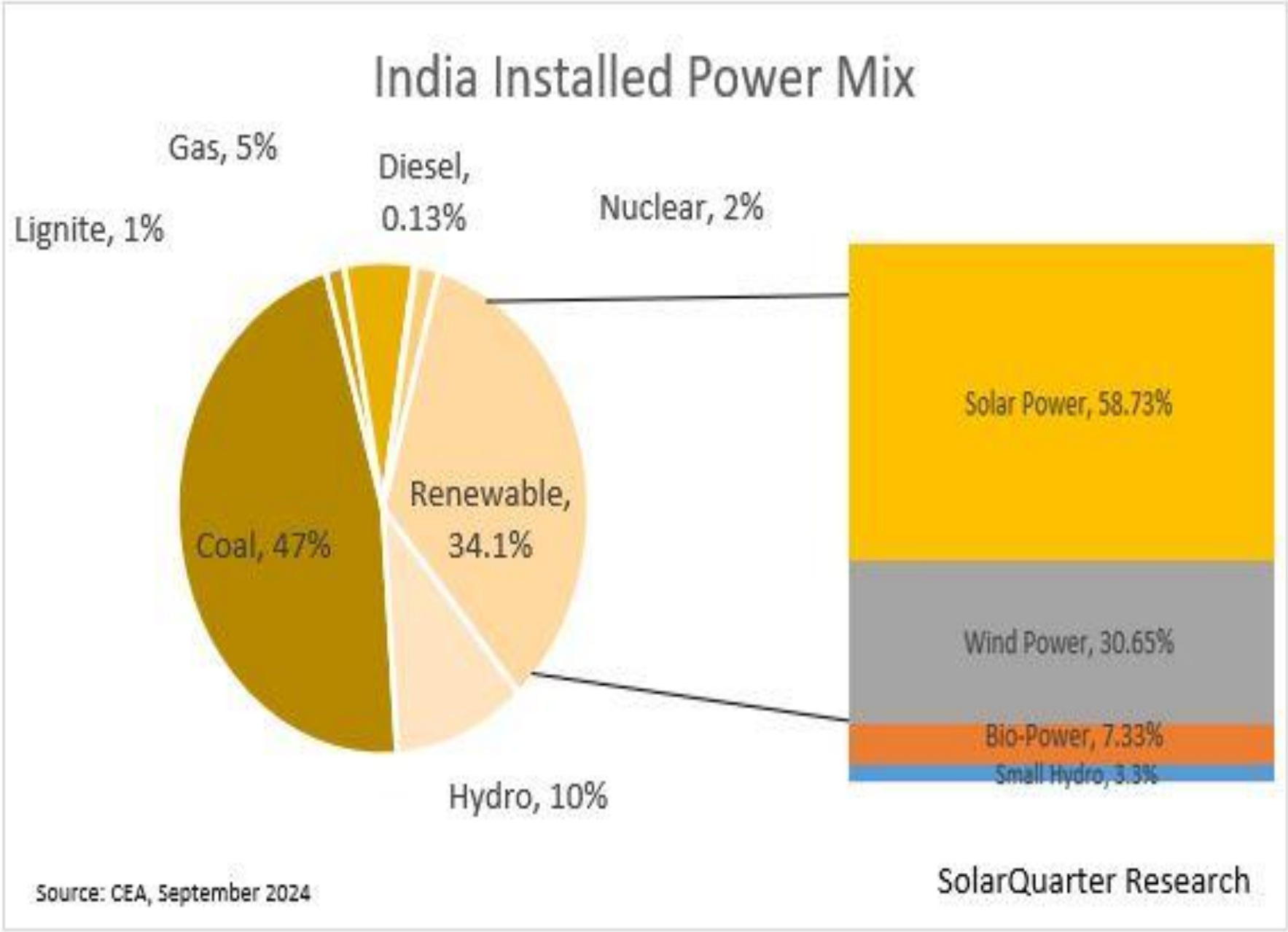


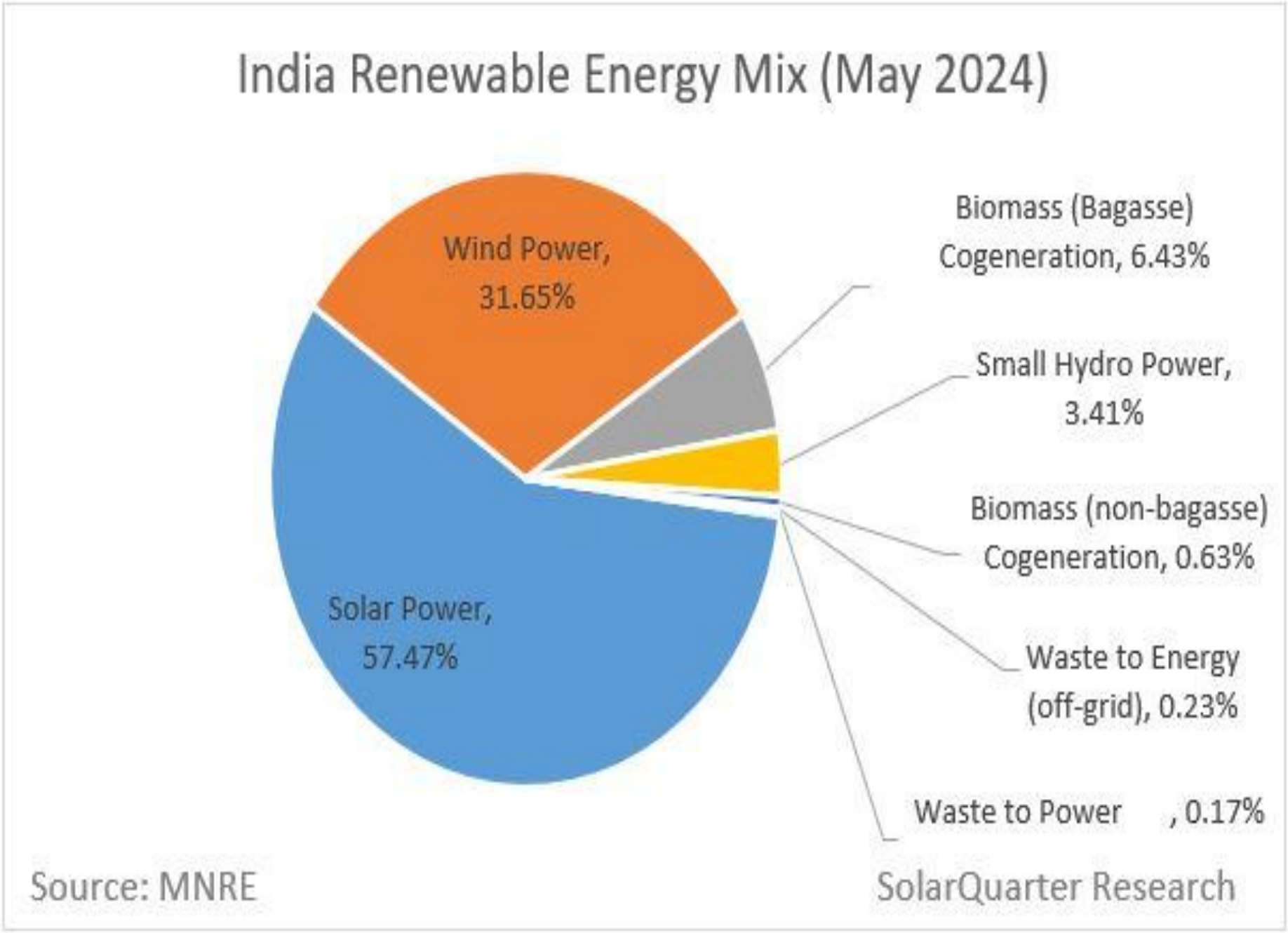
FIG 1:-Total installed Power generation Capacity (end of April 2017)


2018
Stated Policies 2040
Sustainable Development 2040
38%
23%
3%
10%
16%
2%
5%
3%
25%
22%
1%
8%
15%
11%
13%
5%
4%
12%
0.4%
5%
11%
18%
19%
21%
9%
Coal*
Natural gas*
Oil*
CCUS
Nuclear
Hydro
Solar PV
Wind
Other renewables

Energy Consumption in India

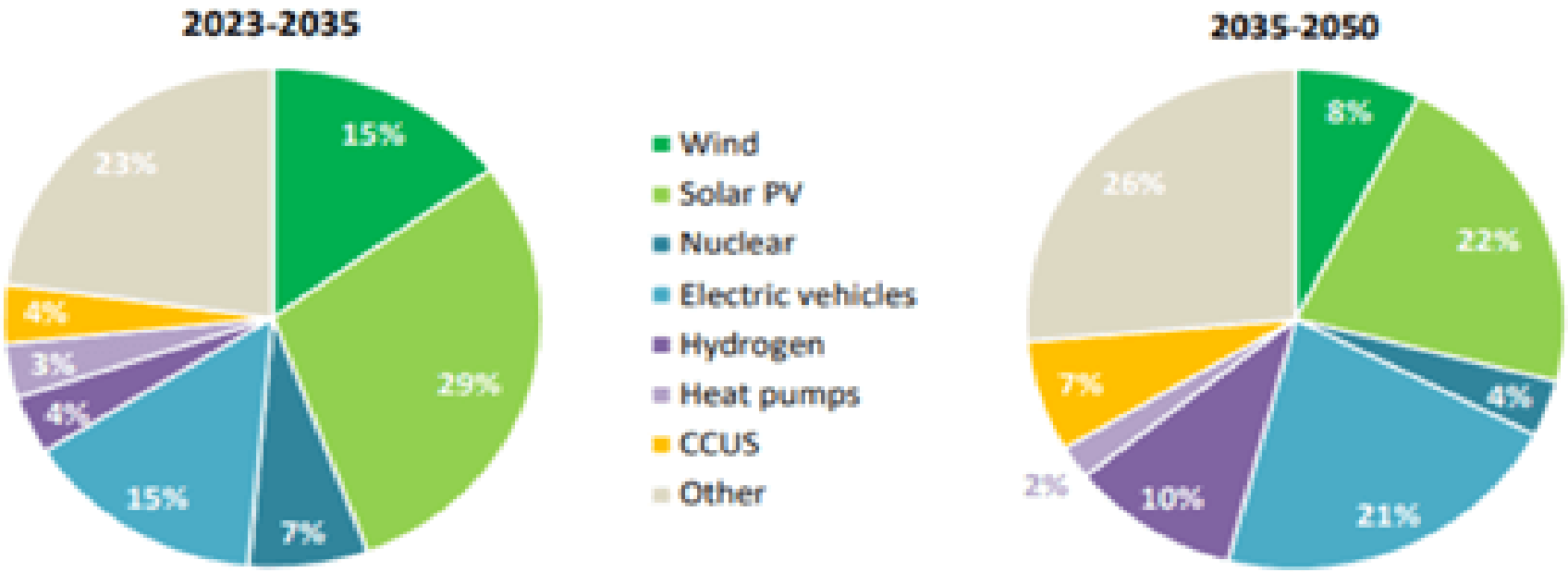



India Installed Power Mix
Gas, 5%
Diesel, 0.13%
Nuclear, 2%
Lignite, 1%
Coal, 47%
Renewable, 34.1%
Hydro, 10%
Solar Power, 58.73%
Wind Power, 30.65%
Bio-Power, 7.33%
Small Hydro, 3.3%
Source: CEA, September 2024
SolarQuarter Research


India Renewable Energy Mix (May 2024)
Wind Power, 31.65%
Solar Power, 57.47%
Biomass (Bagasse) Cogeneration, 6.43%
Small Hydro Power, 3.41%
Biomass (non-bagasse) Cogeneration, 0.63%
Waste to Energy (off-grid), 0.23%
Waste to Power , 0.17%
Source: MNRE
SolarQuarter Research

The global energy scenario

Clean energy technology contribution to energy combustion CO_2 emissions reduction in the APS, 2023-2050

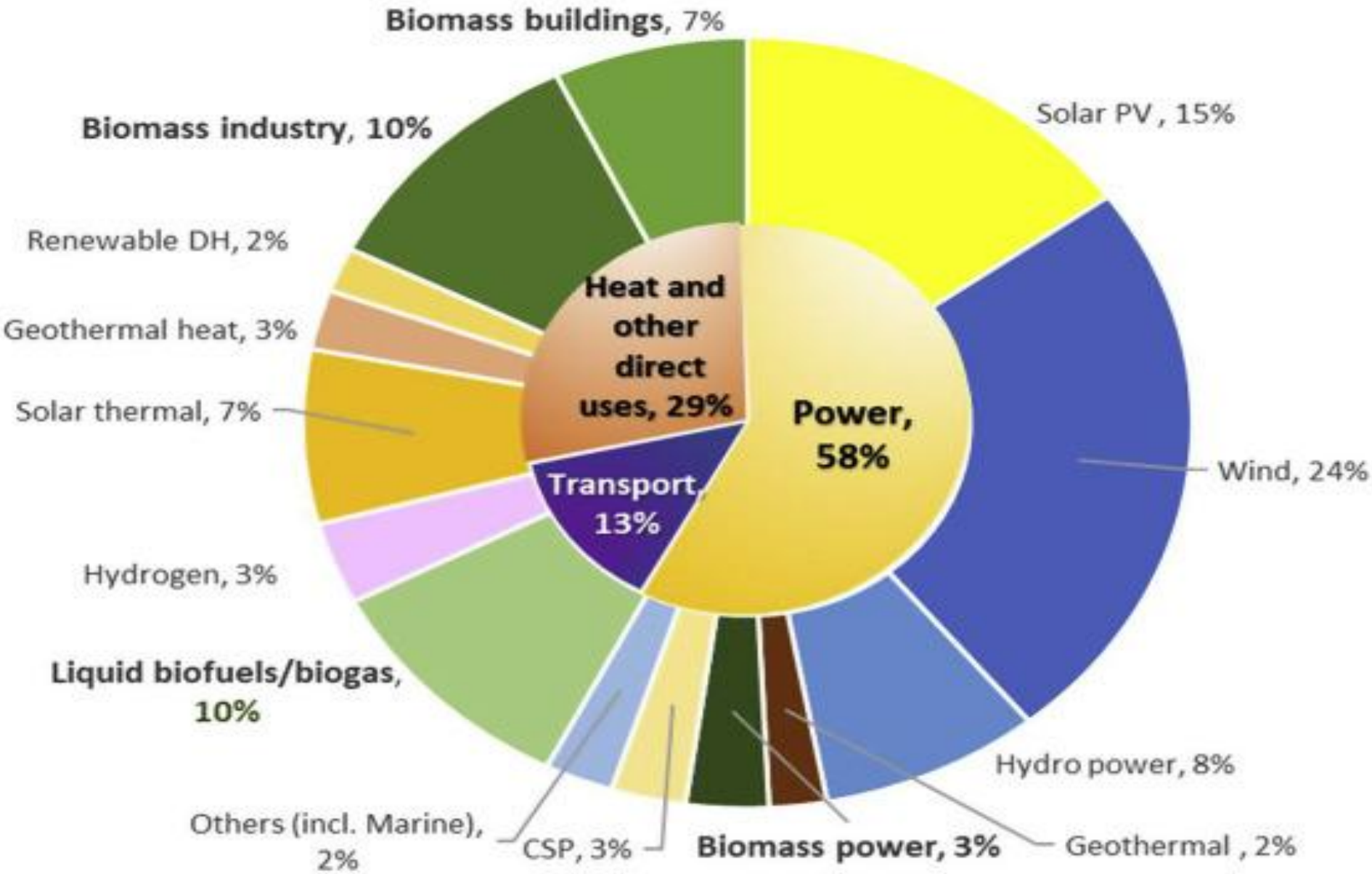


IEA. CC BY 4.0.

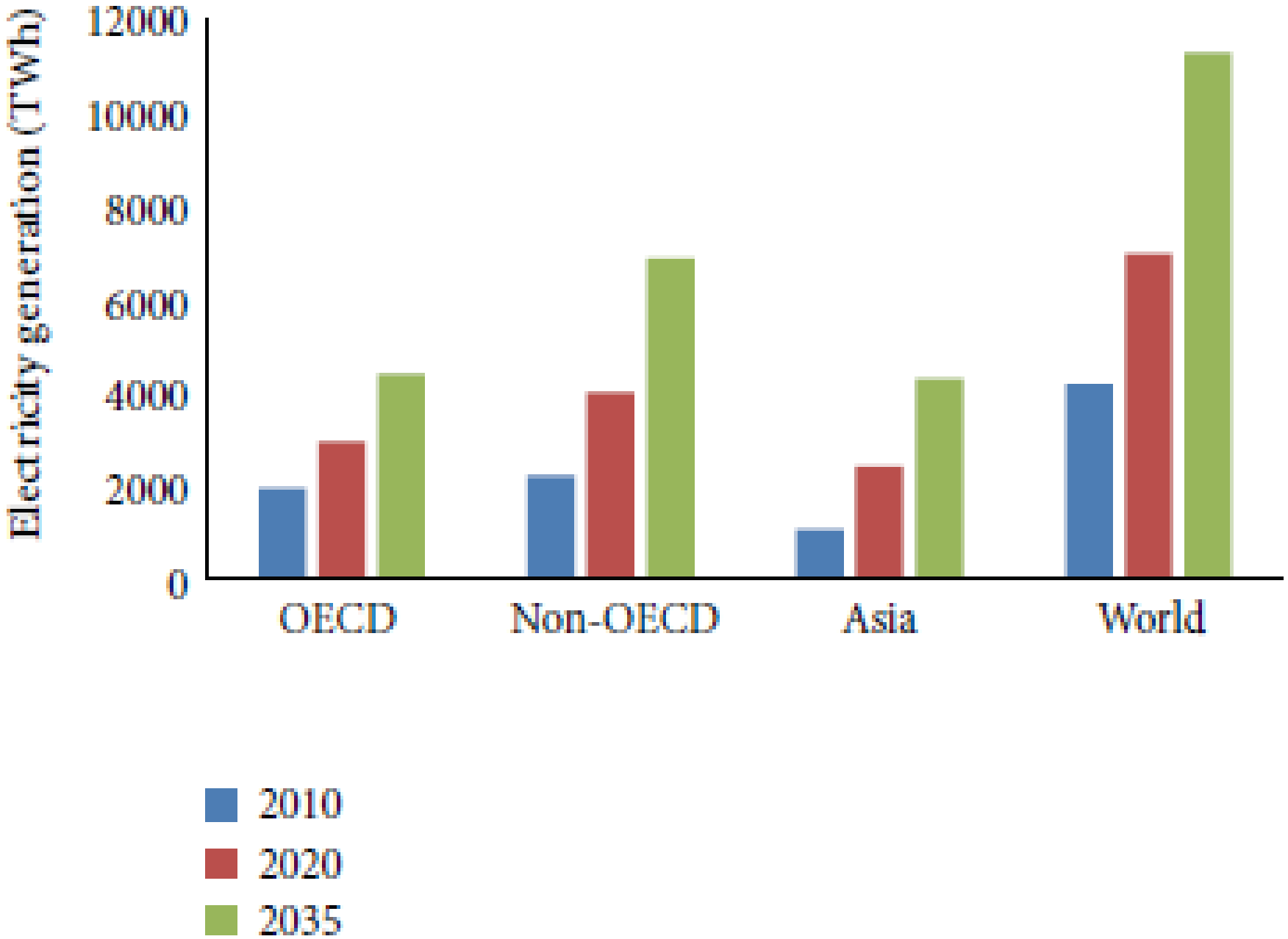
Deployment of seven key clean energy technologies accounts for about three-quarters of CO_2 emissions reductions from today through to 2050

Note: Other includes energy efficiency improvements, other fuel switching such as electrification or switching to biomass, and behaviour changes.

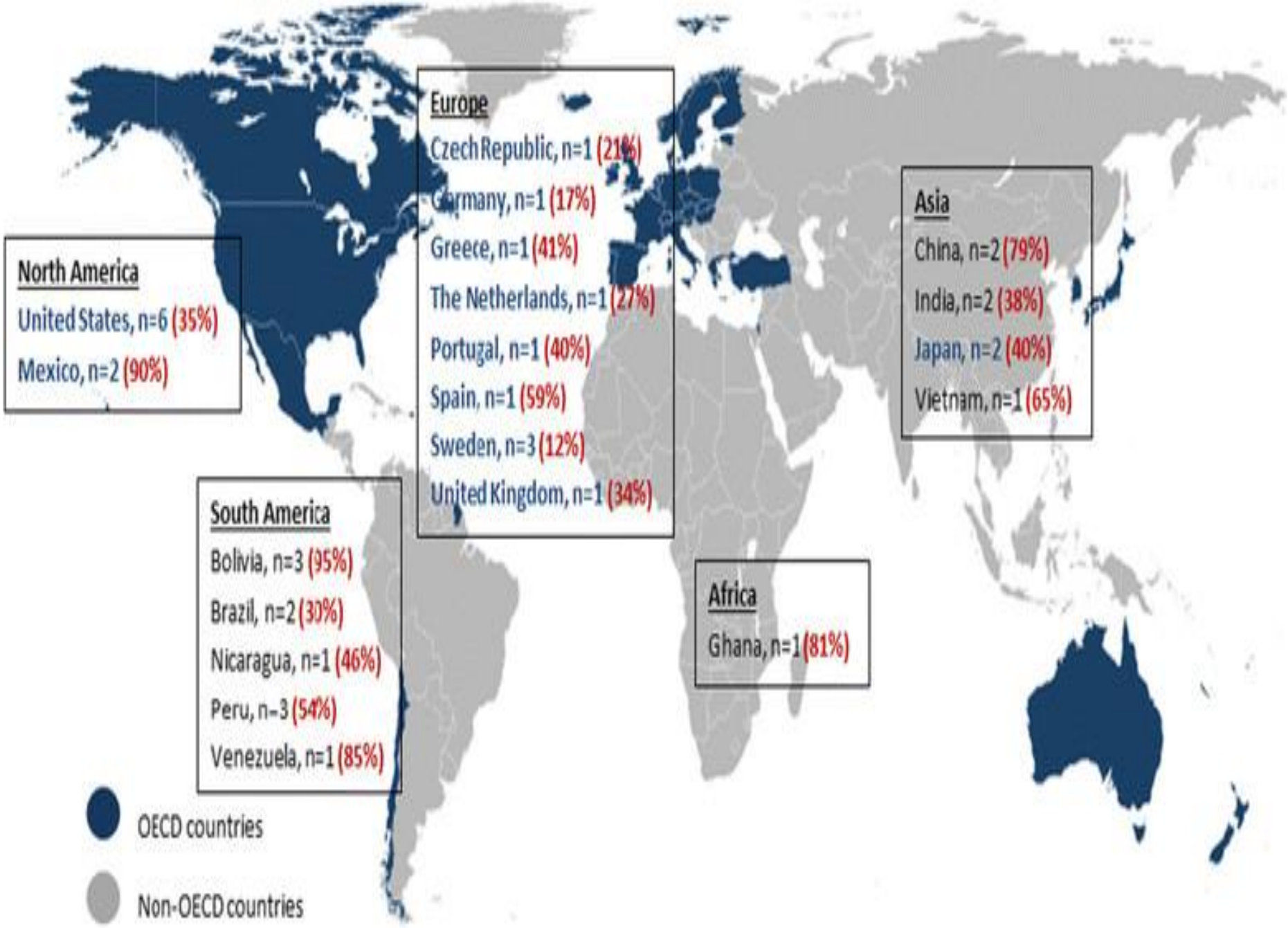
REmap 2050: 222 EJ

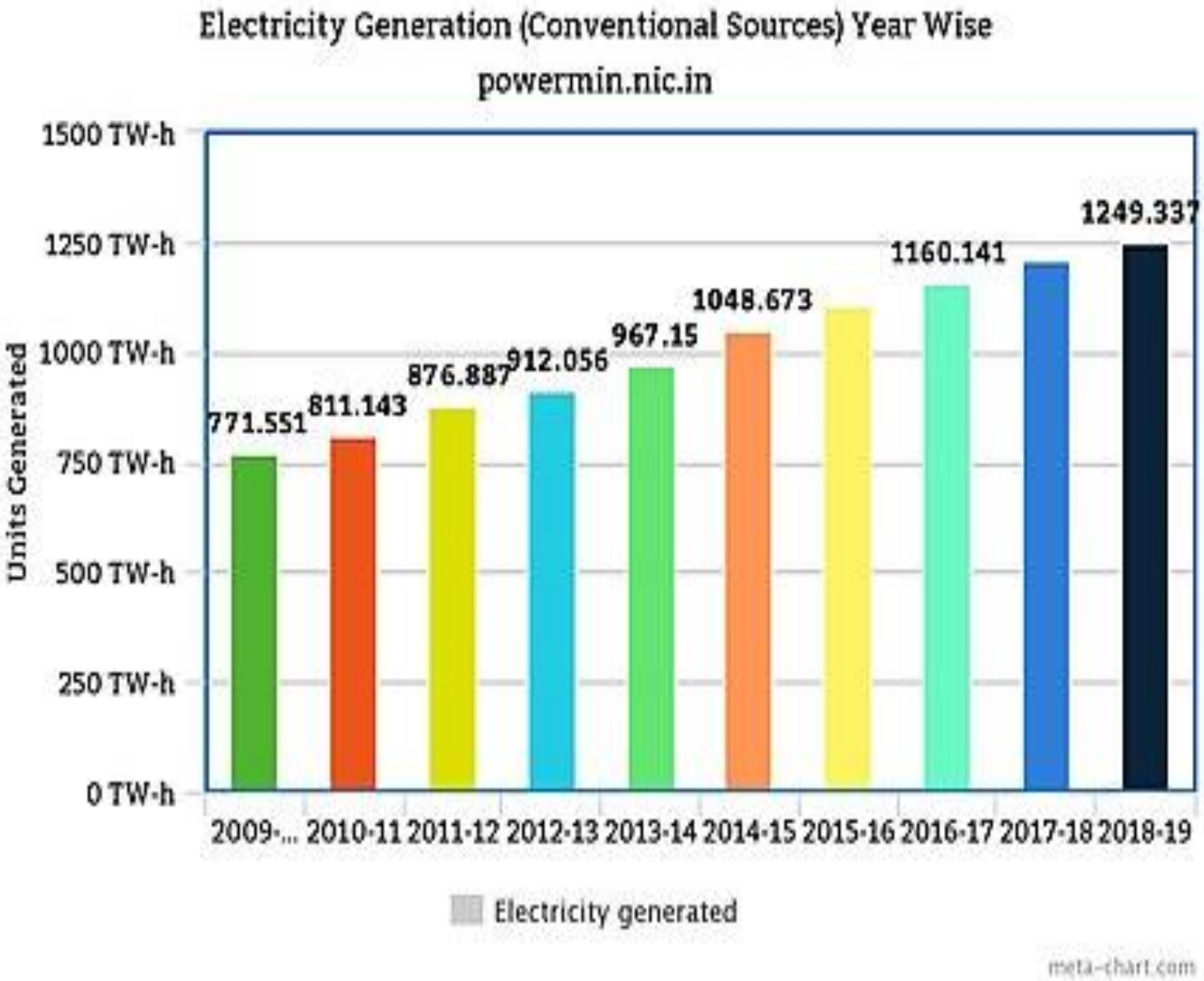


Categorization of energy in different sectors


Electricity generation (TWh)
12000
10000
8000
6000
4000
2000
0
OECD
Non-OECD
Asia
World
2010
2020
2035

- The **Organization for Economic Co-operation and Development (OECD)** is a unique forum where the governments of 38 democracies with market-based economies collaborate to develop policy standards to promote sustainable economic growth.
- The OECD's 38 members are: Austria, Australia, Belgium, Canada, Chile, Colombia, Costa Rica, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Japan, Korea, Latvia, Lithuania, Luxembourg, Mexico, the Netherlands, New Zealand, Norway, Poland, Portugal, Slovak


Europe
Czech Republic, n=1 (21%)
Germany, n=1 (17%)
Greece, n=1 (41%)
The Netherlands, n=1 (27%)
Portugal, n=1 (40%)
Spain, n=1 (59%)
Sweden, n=3 (12%)
United Kingdom, n=1 (34%)
North America
United States, n=6 (35%)
Mexico, n=2 (90%)
Asia
China, n=2 (79%)
India, n=2 (38%)
Japan, n=2 (40%)
Vietnam, n=1 (65%)
South America
Bolivia, n=3 (95%)
Brazil, n=2 (30%)
Nicaragua, n=1 (46%)
Peru, n=3 (54%)
Venezuela, n=1 (85%)
Africa
Ghana, n=1 (81%)
OECD countries
Non-OECD countries


Electricity Generation (Conventional Sources) Year Wise
powermin.nic.in
1500 TW-h
1250 TW-h
1000 TW-h
750 TW-h
500 TW-h
250 TW-h
0 TW-h
Units Generated
771.551
811.143
876.887
912.056
967.15
1048.673
1160.141
1249.337
2009-... 2010-11 2011-12 2012-13 2013-14 2014-15 2015-16 2016-17 2017-18 2018-19
Electricity generated
meta-chart.com

Sector wise Energy Consumption Pattern

Sectors	International Level	National Level	State Level
Industry	51.7%	29%	25%
Transport	26.6%	30%	0.6%
Domestic	13.9%	27%	49%
Commercial	7.8%	9%	17%
Others	0	5%	8.4%

Strategy Energy Partnership (SEP)

- Enhancing Energy Security
- Harnessing Innovation
- Modernizing the Power System
- Enhancing Energy Efficiency and Conservation
- Promoting Energy Trade and Investment
- Promoting Inclusive and Sustainable Economic Growth

- India and the USA have participated in a **virtual ministerial meeting** of the **U.S.-India Strategic Energy Partnership (SEP)** to review progress, highlight major accomplishments, and prioritize new areas for cooperation.

- The SEP was established in **April 2018** whose objective is to encourage meaningful engagements through robust **government-to-government cooperation** and **industry engagement.**

- The SEP organizes **inter-agency engagement** on both sides across **four primary pillars** of cooperation:
    - Power and Energy Efficiency,
    - Oil and Gas,
    - Renewable Energy,
    - Sustainable Growth.

- The SEP also supports USA efforts under the **Asia-EDGE initiative,** which establishes **India as a strong energy partner** in the **Indo-Pacific region.**

- **Major Outcomes:** Number of achievements and priorities for new work under the SEP were announced by both sides. These can be **broadly segregated** under the **four primary pillars of cooperation,** as given below:

1. **Power and Energy Efficiency:**

- **Modernizing the Power System:** Both countries have been collaborating on the integration of **new technologies** into the smart grid; modernizing the power distribution sector i.e. rooftop solar, deployment of Smart Meters etc.

  – The two countries are leading joint research and development (R&D) through **Advance Clean Energy-Research (PACE-R)** on **smart grids and energy storage** to increase resilience and reliability of the electric grid.

- New areas of research on **transformational power generation** based on **supercritical CO_2 (sCO_2) power cycles** and **advanced coal technologies** for power generation including carbon capture, utilization, and storage (CCUS) were also initiated.**sCO_2** is a **fluid state of carbon dioxide** where it is held at or above its critical temperature and critical pressure.
- Commitment to advance **civil nuclear cooperation** i.e. recent progress on the **Westinghouse** (USAs Electric Company) **nuclear reactor project at Kovvada, Andhra Pradesh** were also discussed.

- It also promotes continued bilateral R&D engagement on advanced civil nuclear energy technologies through the **U.S.-India Civil Nuclear Energy Working Group.**

2. Oil and Gas:

- A Memorandum of Understanding (MoU) was signed to begin cooperation on **Strategic Petroleum Reserves operation.**

- The possibility of **India storing oil in the U.S. Strategic Petroleum Reserve** was also discussed.
- Both sides have noted the significant increase in the bilateral hydrocarbon trade through SEP touching 9.2 billion USD during 2019-20 and marking a **93% increase since 2017-18,**

  – Through the **U.S.-India Natural Gas Task Force,** it affirmed to promote greater hydrocarbon trade between the two countries.

3. Renewable Energy:

- The both sides launched a **public-private Hydrogen Task Force** to help scale up technologies to **produce hydrogen from renewable energy and fossil fuel sources** and to bring down the cost of deployment for enhanced energy security and resiliency.
- **Efficient Buildings and Clean Technologies:** A MoU was signed to collaborate on India's first-ever **Solar Decathlon in 2021,** establishing a **collegiate competition** to prepare the next generation to design and build high efficiency buildings powered by renewables.

  – In response to the Covid-19 **pandemic**, **United States Agency for International Development (USAID)** and **Energy Efficiency Services Ltd (EESL)**, India has jointly initiated **"Retrofit of Air Conditioning to Improve Air Quality for Safety and Efficiency" (RAISE)** for healthy and energy efficient buildings.

  – Launched collaboration between the **USA Department of Energy (DOE) National labs** and the **Indian National Institutes under the Ministry of New & Renewable Energy** as part of the **South Asia Group for Energy (SAGE), supported by USAID,** for joint research on development of advanced clean technologies.

- Information exchange on **sustainable biofuel production (bioethanol and renewable diesel)** and use for air and sea transport were also discussed.
- It will also explore cooperation on utilizing the economic value of converting bio-waste into biogas.

4. Sustainable Growth:

- The best practices are being adopted through **methodologies in energy data management and capacity building in energy modeling.**

- **USAID** and **NITI Aayog** jointly launched the **India Energy Modeling Forum** to build a network of modeling communities and its linkage with Government for analytical work and policy making exercise.

Classification of Energy

- Primary Energy sources (natural gas, crude oil)
- Secondary Energy sources (refined crude oil products)
- Commercial Energy sources (electricity)
- Non-commercial Energy sources (firewood)
- Renewable Energy sources
- Non-renewable Energy sources

Primary Energy sources

Primary energy is the energy that's harvested directly from natural resources.

Sources of primary energy fall into two basic categories, fuels and flows.

The fuels in primary energy are all primary fuels.

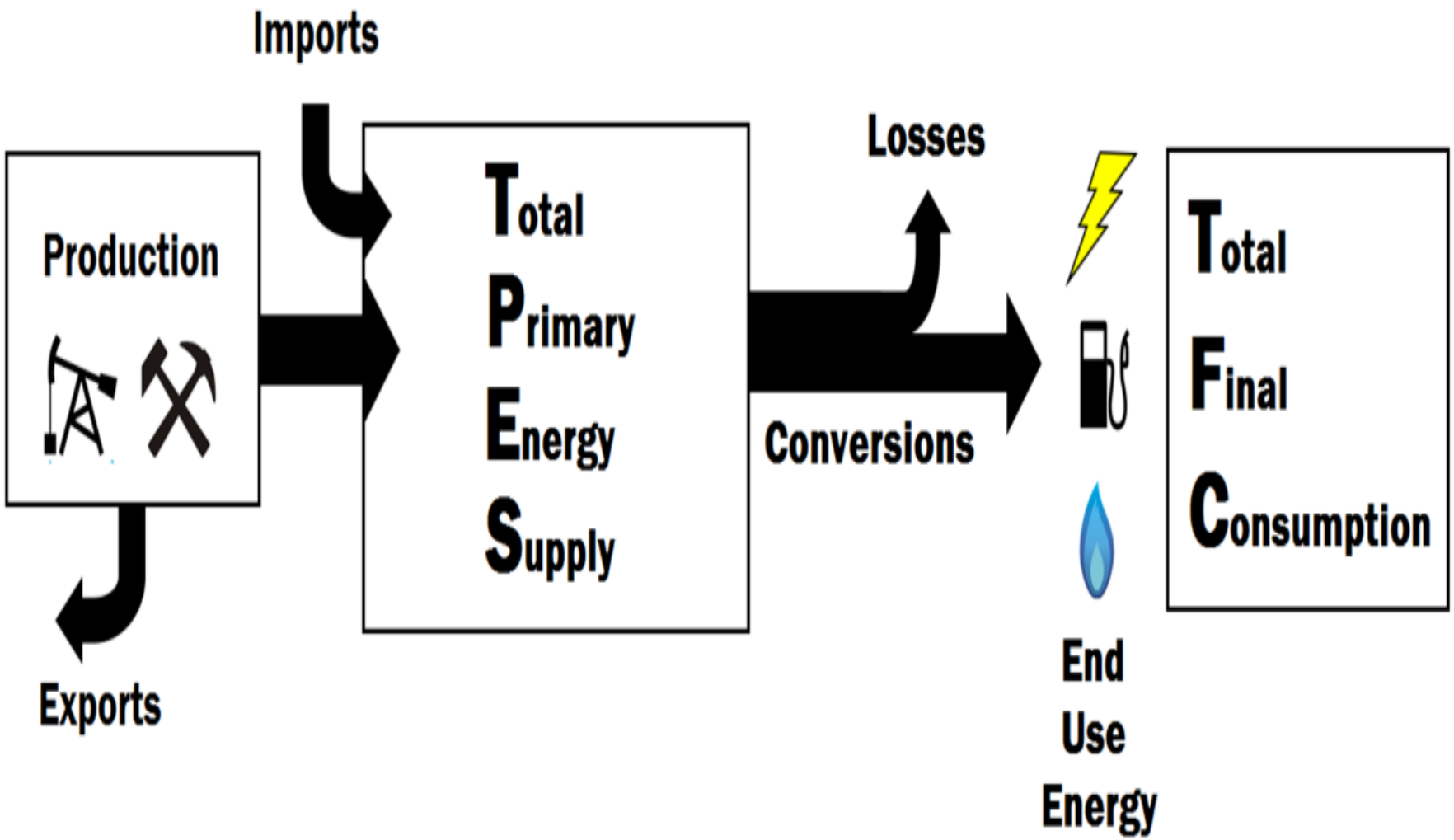
A country's different sources of primary energy are aggregated into a quantity called **total primary energy supply (TPES).**

All of human energy must come from one of these primary energy sources, there are no energy alternatives. **Primary energy is contrasted with end use energy.** Primary energy almost always needs to be converted through an energy conversion technology to make this primary energy source into an energy currency or a secondary fuel before it can be used.

For example

- Coal is usually put into a coal-fired power plant to generate electricity.
- Wind must be harnessed by a wind turbine before it can generate electricity.

Country's Energy Flows



Secondary Energy sources

Secondary energy sources are derived from primary sources in a from of either final fuel or energy supply.

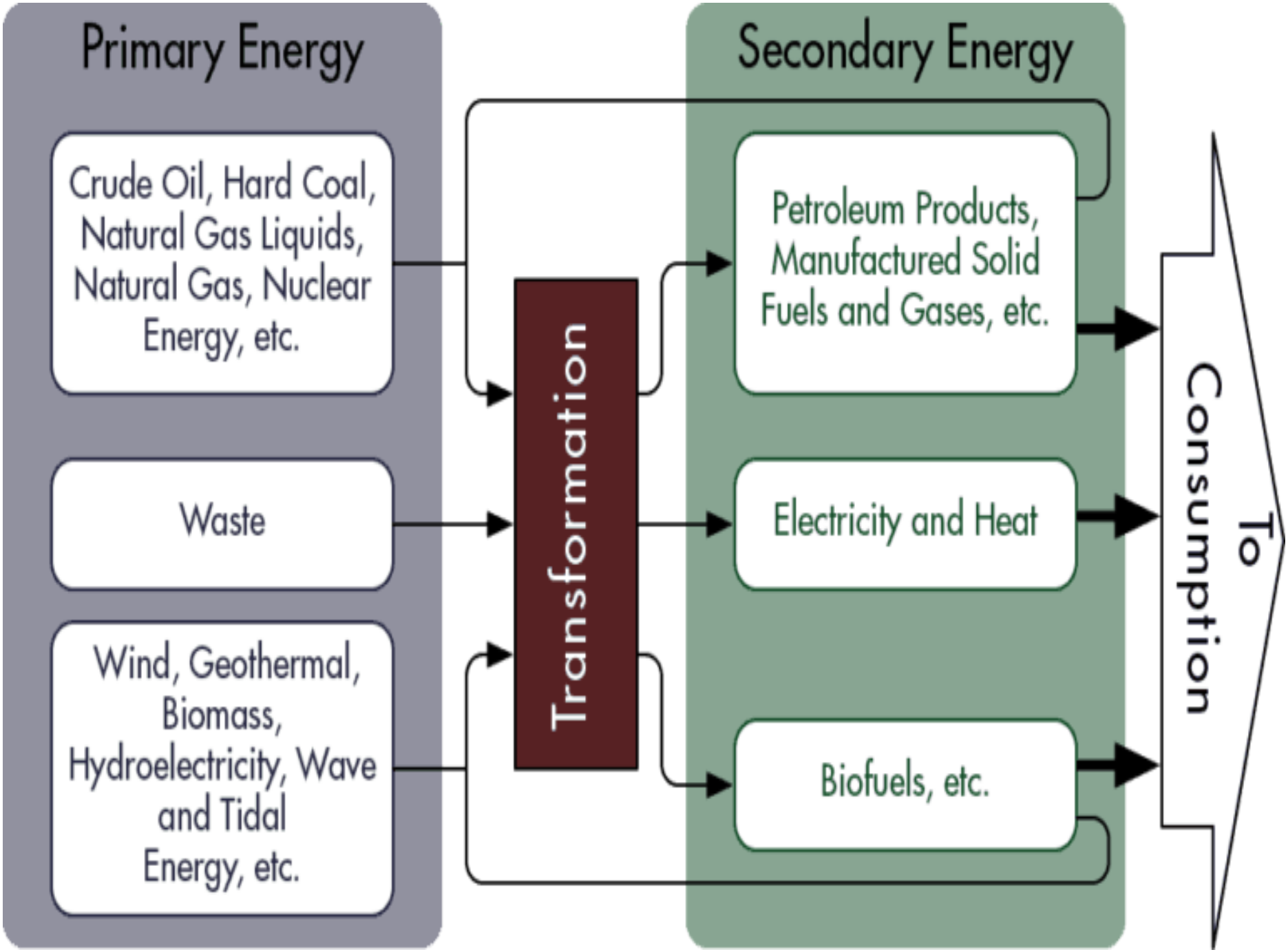
Involvement of technological processes in this transformation in between causes drop in primary energy on the way to consumers.

Secondary energy sources are also referred to as energy carriers, because they move energy in a useable form from one place to another.

The two most well-known energy carriers are: **Electricity , Hydrogen**

Example

- Crude oil is refined and converted in products like petrol, diesel, kerosene.


Primary Energy
Crude Oil, Hard Coal, Natural Gas Liquids, Natural Gas, Nuclear Energy, etc.
Waste
Wind, Geothermal, Biomass, Hydroelectricity, Wave and Tidal Energy, etc.
Transformation
Secondary Energy
Petroleum Products, Manufactured Solid Fuels and Gases, etc.
Electricity and Heat
Biofuels, etc.
To Consumption

Commercial Energy sources

The energy sources that are available in the market for a definite price are known as commercial energy.

By far the most important form of commercial energy are electricity, coal and refined petroleum products.

Commercial energy forms the basis of industrial, agriculture, transport, and commercial development in modern world. In industrialized countries, commercialized fuels are predominant sources not only for economic production, but also for many household tasks of general population.

Examples

Electricity, Lignite, Coal, oil etc.

Non-commercial Energy sources

The energy sources that are not available in the market for a price are known as non-commercial energy sources.

Non-commercial energy sources include fuels such as **firewood, cattle dung and agriculture waste,** which are traditionally gathered, and not bought at a price used especially in households. These are also called as **traditional fuels.** Non-commercial energy is often ignored in energy accounting.

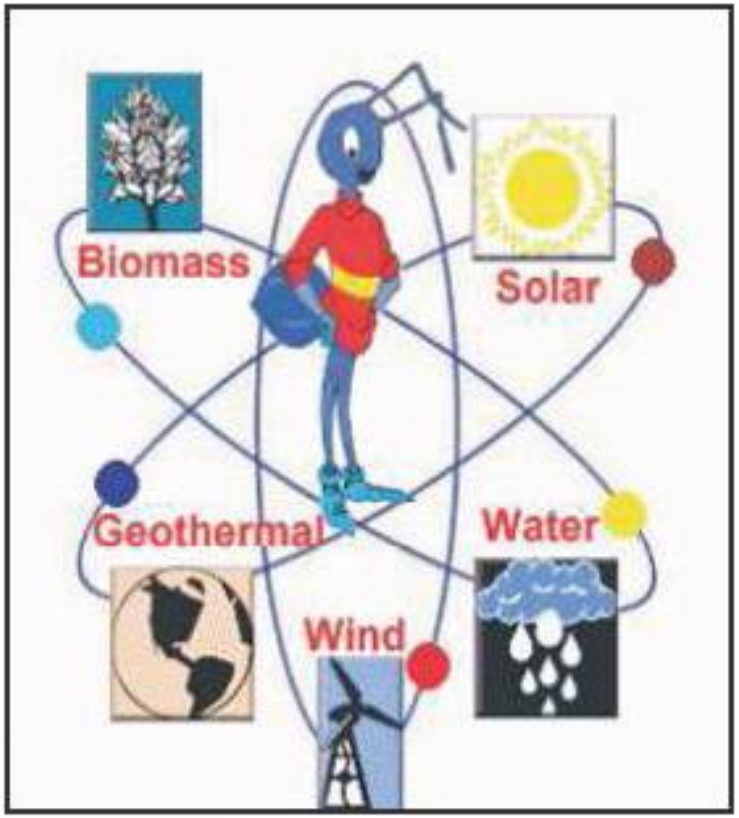
Example

Firewood, agro waste from rural areas.

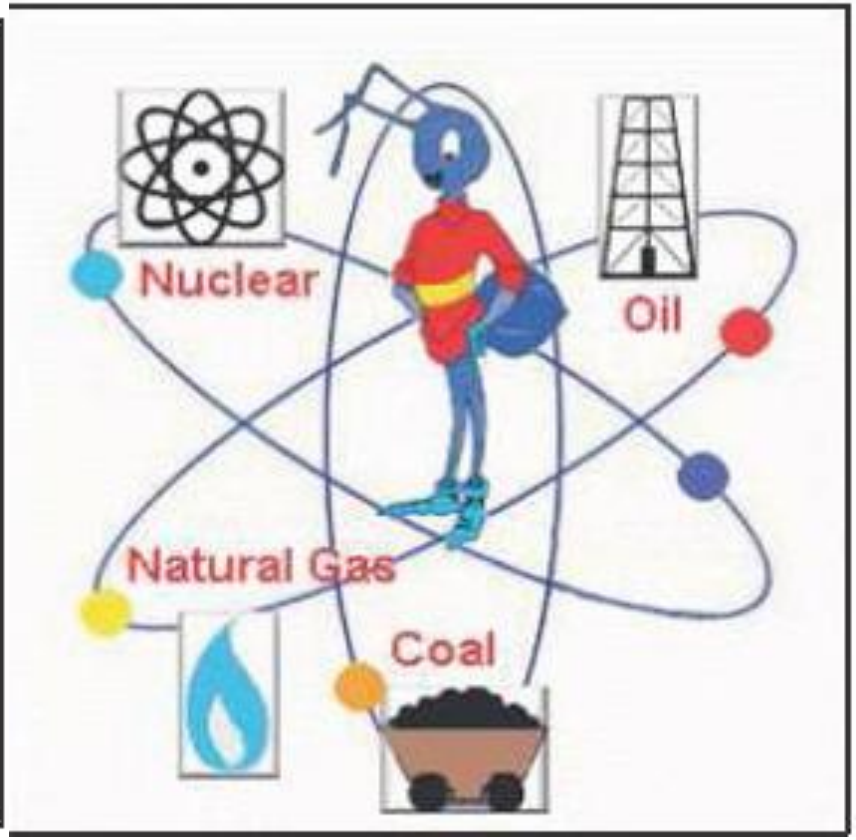
Differences between Renewable and Non-renewable Resources

Renewable Resources	Non-renewable Resources
Depletion	
Renewable resources cannot be depleted over time	Non-renewable resources deplete over time
Sources	
Renewable resources include sunlight, water, wind and also geothermal sources such as hot springs and fumaroles	Non-renewable energy includes fossil fuels such as coal and petroleum.
Environmental Impact	
Most renewable resources have low carbon emissions and low carbon footprint	Non-renewable energy has a comparatively higher carbon footprint and carbon emissions.
Cost	
The upfront cost of renewable energy is high. – For instance, Generating electricity using technologies running on renewable energy is costlier than generating it with fossil fuels	Non-renewable energy has a comparatively lower upfront cost.
Infrastructure Requirements	
Infrastructure for harvesting renewable energy is prohibitively expensive and not easily accessible in most countries.	Cost-effective and accessible infrastructure is available for non-renewable energy across most countries
Area Requirements	
Requires a large land/ offshore area, especially for wind farms and solar farms	Comparatively lower area requirements

Renewable Energy



Non-Renewable energy



Renewable Energy sources

This is the energy acquired from never ending sources of energy available in nature.

The main feature of this is, it can be extracted without causing pollution.

Renewable energy is energy that is collected from renewable resources, which are naturally replenished on a human timescale, such as **sunlight, wind, rain, tides, waves, and geothermal heat.**

Renewable energy often provides energy in four important areas: electricity generation, air and water heating/cooling, transportation, and rural (off-grid) energy services.

Example

Solar energy, wind energy, tidal energy ,hydo energy, bio mass, geothermal energy,

Non-renewable Energy sources

Non renewable energy is the energy obtained from the conventional fuels which are exhaustible today or tomorrow with time.

A non-renewable resource (also called a finite resource) is a resource that does not renew itself at a sufficient rate for sustainable economic extraction in meaningful human time-frames. An example is carbon-based, organically-derived fuel. The original organic material, with the aid of heat and pressure, becomes a fuel such as oil or gas. Earth minerals and metal ores, fossil fuels (coal, petroleum, natural gas) and groundwater in certain aquifers are all considered non-renewable resources, though individual elements are always conserved.

Example

Coal, oil, gas, Hydro power, Diesel power.

Renewable energy and Non-renewable energy

- Renewable energy is energy obtained from sources that are essentially inexhaustible.

- Examples of renewable resources include wind power, solar power, geothermal energy, tidal power and hydroelectric power.

- The most important features of renewable energy is that it can be harnessed without the release of harmful pollutants.

- Non-renewable energy is the conventional fossil fuels such as coal, oil and gas, which are likely to deplete with time.

Fossil fuels

Adverse effects of fossil fuels on environment Greenhouse gases and Global warming

- https://en.wikipedia.org/wiki/Fossil_fuel
- https://www.eesi.org/topics/fossil-fuels/description
- https://www.aplustopper.com/different-types-of-fossil-fuels/
- https://www.inspirecleanenergy.com/blog/clean-energy-101/advantages-disadvantages-fossil-fuels

What are Fossil Fuels?

- Fossil fuels are buried flammable geologic deposits of organic substances such as dead plants, and animals that got deposited under several thousand feet of silt.

- These deposits decayed with the passage of time and got converted to natural gas, coal, and petroleum due to the extreme heat and pressure inside the earth's crust.

- They are also known as non-renewable sources of energy as it takes a very long time for it to replenish.

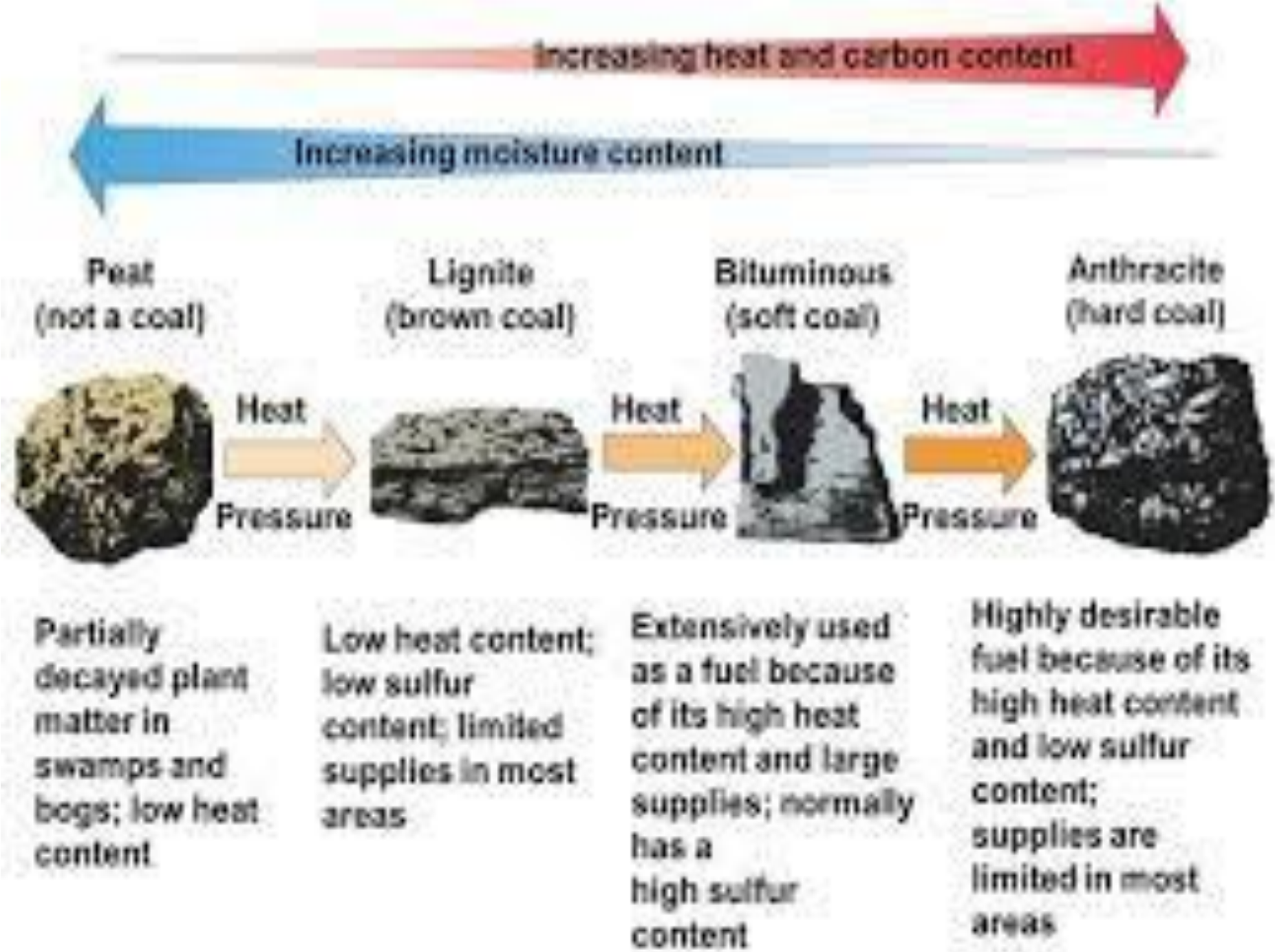
Types, Formation and Uses of Fossil Fuels

Fossil fuels are of the following types:

- Coal
- Petroleum
- Natural gas

Coal

- It is a hard, black coloured substance made up of carbon, hydrogen, nitrogen, oxygen, and sulphur.
- The major types of coal are- **anthracite, bituminous and lignite**.
- **Anthracite** has a higher carbon concentration and is the hardest type of coal.
- **Lignite** has a high concentration of oxygen and hydrogen but a low concentration of carbon.
- **Bituminous** is a moderate form of coal.
- Coal is processed industrially to obtain derivatives like coke, coal tar, and coal gas.


Increasing heat and carbon content
Increasing moisture content
Peat
(not a coal)
Lignite
(brown coal)
Bituminous
(soft coal)
Anthracite
(hard coal)
Heat
Pressure
Heat
Pressure
Heat
Pressure
Partially decayed plant matter in swamps and bogs; low heat content
Low heat content; low sulfur content; limited supplies in most areas
Extensively used as a fuel because of its high heat content and large supplies; normally has a high sulfur content
Highly desirable fuel because of its high heat content and low sulfur content; supplies are limited in most areas

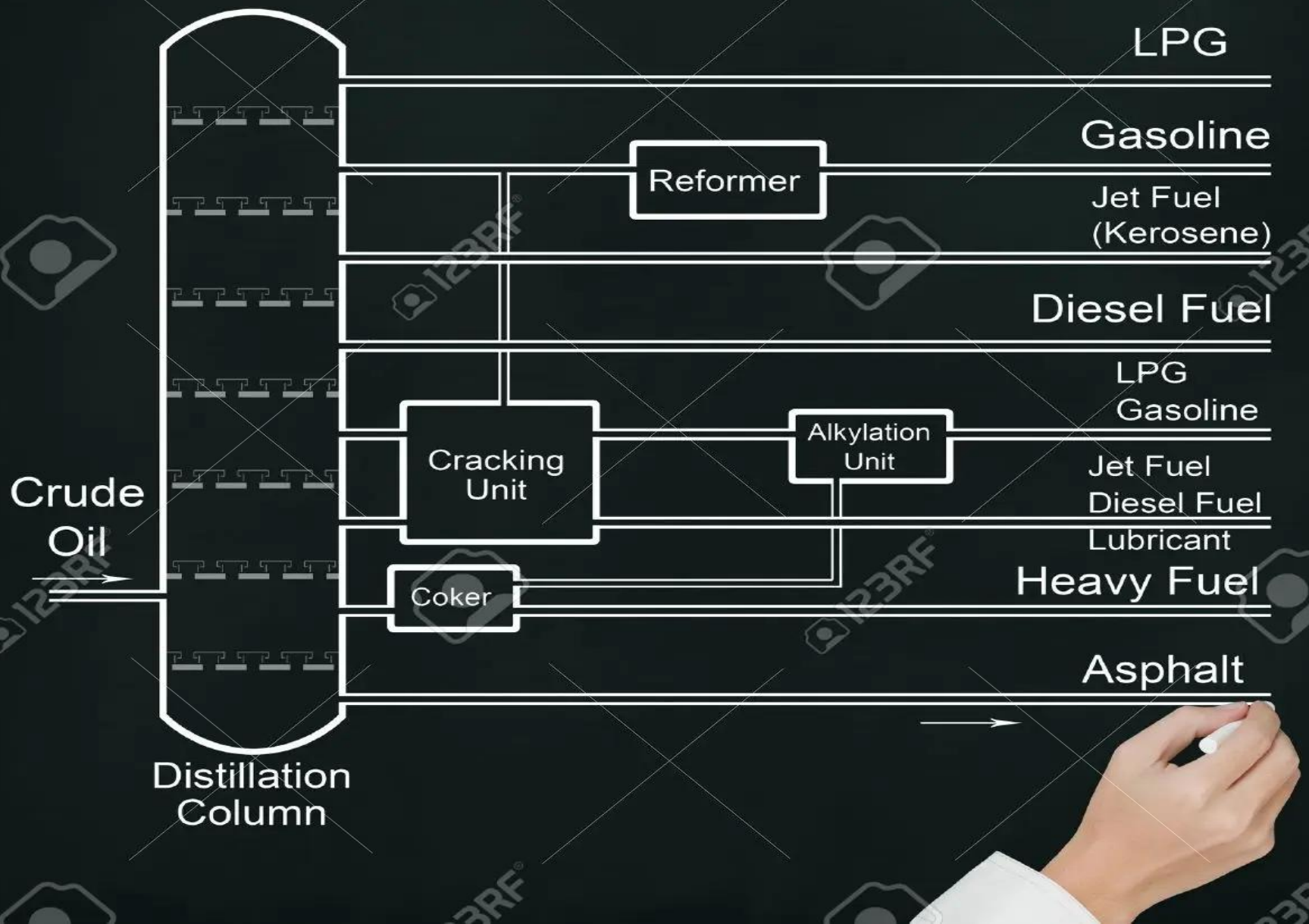
- **Formation of Coal**
- The process of formation of coal is known as coalification.
- The dense forest present in the low-lying wetland got buried in the earth, millions of years ago.
- Soil kept depositing over them and they got compressed.
- As they went deeper and deeper, they faced high temperature and pressure.
- As a result, the substances slowly got converted into coal.

- **Uses of Coal**
- Coal was used to produce steam in the railway engines initially.
- It is used to cook food.
- It is used to generate electricity in thermal plants.
- It is used in industries as fuel.

Petroleum

- It is a clear, oily liquid, usually green or black in colour.
- It has a very strange smell and is a mixture of petroleum gas, diesel, paraffin wax, petrol, lubricating oil, etc.
- It is also termed as "**Black Gold**" because of its wide range of uses in many industries.

Refinery


LPG
Gasoline
Reformer
Jet Fuel
(Kerosene)
Diesel Fuel
LPG
Gasoline
Alkylation
Unit
Cracking
Unit
Jet Fuel
Diesel Fuel
Lubricant
Crude
Oil
Coker
Heavy Fuel
Asphalt
Distillation
Column

- **Formation of Petroleum**

- The sea animals and plants died and their bodies settled at the bottom of the sea.
- They got compressed by the layers of sand and clay.
- Their encounter with high temperature and pressure converts them into petroleum.
- The petroleum is separated from the crude oil by a series of processes in a refinery. This is known as petroleum refining.

- **Uses of Petroleum**

- It is used to power internal combustion engines in the form of petrol.
- It is used in roofing, road pavements, and as a water repellent.
- It is used in manufacturing detergents, plastics, fibers, polyethene, etc.

Natural gas

- It is a clean and non-toxic fossil fuel.
- It is colorless and odorless and can be easily transferred through pipelines.
- It is stored as compressed natural gas (CNG) under high pressure.
- It is a less polluting and less expensive fossil fuel.
- Methane is the most important natural gas.

Formation of Natural Gas

- The phytoplankton and zooplankton sink to the bottom of the ocean and mix with organic materials to form an organic-rich mud.
- The mud buried under more sediments and lithifies to form an organic shale. This prevents its exposure to oxygen. This is done to protect the organic materials from being decomposed by bacteria.
- The increasing pressure and temperature transform the shale into a waxy material known as the kerogen.
- At temperatures between 90-160°C kerogen is transformed into natural gas.

Uses of Natural gas

- Compressed Natural Gas is used for generating power.
- It is used as fuels in automobiles.
- It can be used at homes for cooking.
- It is used as a starting material in chemicals and fertilizers.

Advantages and Disadvantages of Fossil Fuels

Advantages:

- Fossil fuels can generate a large amount of electricity at a single location.
- They can be found very easily.
- They are cost-effective.
- Transportation of oil and gas can be done easily through pipelines.
- They have become safer over time.
- Despite being a finite resource, it is available in plenty.

Disadvantages

- Fossil fuels emit carbon dioxide when burnt which is a major greenhouse gas and the primary source of pollution.
- This has contributed to global warming.
- They are a non-renewable resource, i.e., once used they cannot be replaced.
- Combustion of fossil fuels makes the environment more acidic.
- This has led to unpredictable and negative changes in the environment.
- Harvesting of fossil fuels also causes fatal diseases among the people. For example the coal miners often suffer from Black Lung Disease.
- The natural gas drillers are constantly exposed to chemicals and silica which is dangerous for their health.

Green house gases and global warming.

- https://www.nrdc.org/stories/global-warming-101
- https://en.wikipedia.org/wiki/Greenhouse_effect
- https://www.joboneforhumanity.org/what_is_global_warming_sign_up?gclid=EAIaIQobChMIk8GkhcbI7wIV0ZVLBR1V9gSHEAAYAiAAEgJ4I_D_BwE
- https://www.britannica.com/science/greenhouse-gas
- https://warmheartworldwide.org/climate-change/?gclid=EAIaIQobChMIvdnBo8bI7wIVrJ1LBR2rBwKnEAAYASAAEgKavPD_BwE

Green House effect

- The greenhouse effect is the way in which heat is trapped close to the surface of the Earth by "greenhouse gases."
- These heat-trapping gases can be thought of as a blanket wrapped around the Earth, which keeps it toastier than it would be without them.
- Greenhouse gases include carbon dioxide, methane , nitrous oxides , fluorinated gases and water vapour.

- Greenhouse gases arise naturally, and are part of the make-up of our atmosphere.

- Earth is sometimes called the “Goldilocks” planet – it’s not too hot, not too cold, and the conditions are just right to allow life, including us, to flourish.

- Part of what makes Earth so amenable is the naturally-arising greenhouse effect, which keeps the planet at a friendly 15 °C (59 °F) on average.

- But in the last century or so, humans have been interfering with the energy balance of the planet, mainly through the burning of fossil fuels that give off additional carbon dioxide into the air.

- The level of carbon dioxide in Earth’s atmosphere has been rising consistently for decades and traps extra heat near the surface of the Earth, causing temperatures to rise.

How green house effect increases the temperature

- A greenhouse is a building with glass walls and a glass roof. Greenhouses are used to grow plants, such as tomatoes and tropical flowers.

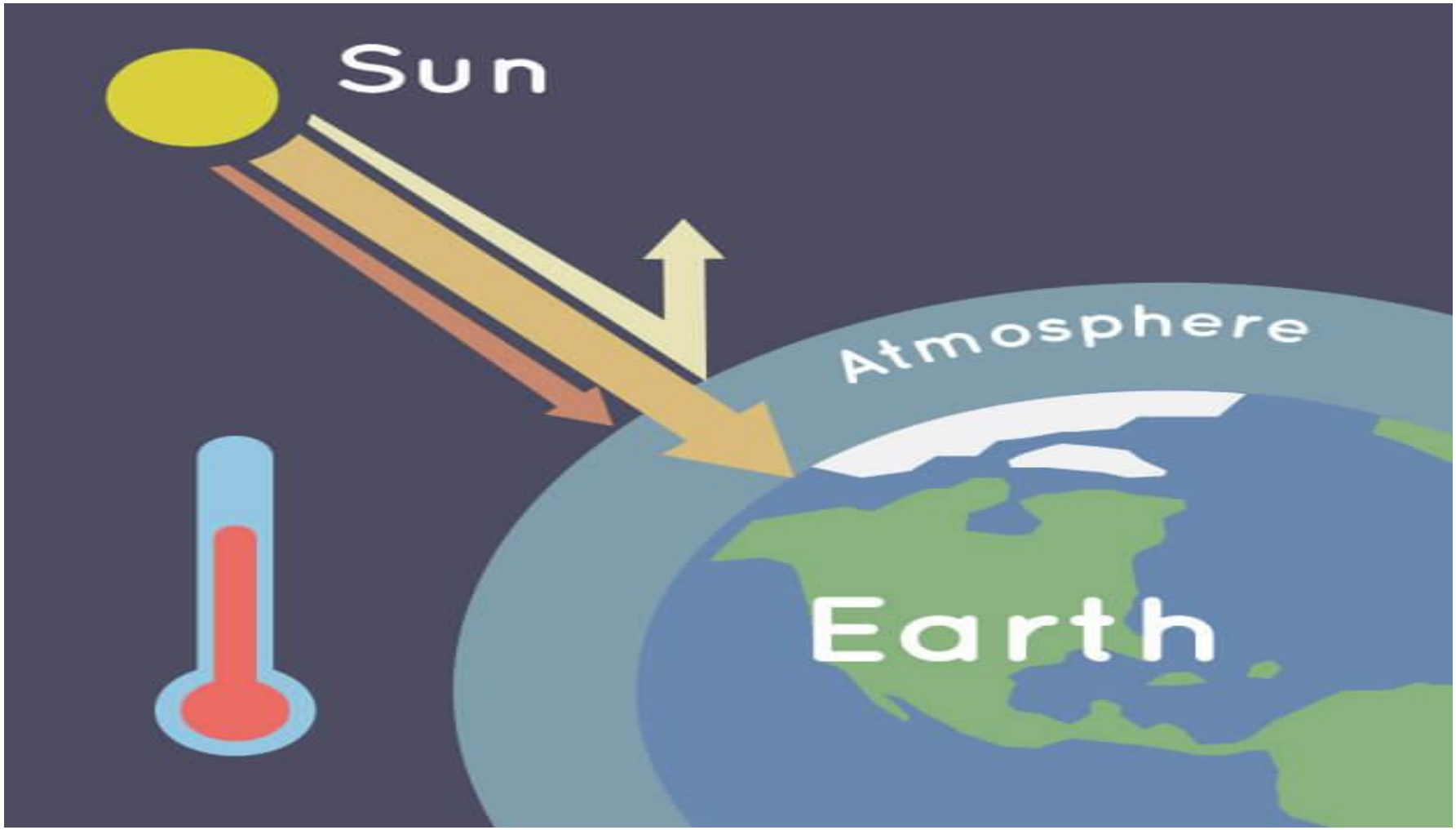
- A greenhouse stays warm inside, even during the winter. In the daytime, sunlight shines into the greenhouse and warms the plants and air inside.

- At nighttime, it's colder outside, but the greenhouse stays pretty warm inside. That's because the glass walls of the greenhouse trap the Sun's heat.

- The greenhouse effect works much the same way on Earth.

- Gases in the atmosphere, such as carbon dioxide, trap heat just like the glass roof of a greenhouse.
- These heat-trapping gases are called greenhouse gases.
- During the day, the Sun shines through the atmosphere. Earth's surface warms up in the sunlight.
- At night, Earth's surface cools, releasing heat back into the air.
- But some of the heat is trapped by the greenhouse gases in the atmosphere.
- That's what keeps our Earth a warm and cozy 58 degrees Fahrenheit (14 degrees Celsius), on average.

- Human activities are changing Earth's natural greenhouse effect.
  Burning fossil fuels like coal and oil puts more carbon dioxide into our atmosphere.



Green house Gases

- ***Carbon Dioxide***
- Accounting for about 76 percent of global human-caused emissions, carbon dioxide(CO) sticks around for quite a while.
- Once it’s emitted into the atmosphere, 40 percent still remains after 100 years, 20 percent after 1,000 years, and 10 percent as long as10,000 years later.

- ***Methane***
- Although methane (CH_4) persists in the atmosphere for far less time than carbon dioxide (about a decade), it is much more potent in terms of the greenhouse effect.
- In fact, pound for pound, its global warming impact is 25 times greater than that of carbon dioxide over a 100-year period.
- Globally it accounts for approximately 16 percent of human-generated greenhouse gas emissions.

- ***Nitrous Oxide***
- Nitrous oxide (NO) is a powerful greenhouse gas.
- It has a GWP 300 times that of carbon dioxide on a 100-year time scale, and it remains in the atmosphere, on average, a little more than a century.
- It accounts for about 6 percent of human-caused greenhouse gas emissions worldwide.

Fluorinated Gases

- Emitted from a variety of manufacturing and industrial processes, fluorinated gases are man-made.

- There are four main categories:
  - Hydro fluorocarbons (HFCs),
  - Per fluorocarbons (PFCs),
  - sulfur hexafluoride (SF_6), and
  - nitrogen tri fluoride (NF_3).

- The green house gasses generated in the environment are just 2% of man-made global greenhouse gas emissions, they trap substantially more heat.

- Indeed, the GWP for these gases can be in the thousands to tens of thousands, and they have long atmospheric lifetimes, in some cases lasting tens of thousands of years.

- HFCs are used as a replacement for ozone-depleting chlorofluorocarbons (CFCs) and hydro chlorofluorocarbons (HCFCs), usually in air conditioners and refrigerators, but some are being phased out because of their high GWP.

- Replacing these HFCs and properly disposing of them is considered to be one of the most important climate actions the world can take.

GWP(Global Warming Potentials)

- It is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO_2).

- The larger the GWP, the more that a given gas warms the Earth compared to CO_2 over that time period.

- The time period usually used for GWPs is 100 years.

- GWPs provide a common unit of measure, which allows analysts to add up emissions estimates of different gases (e.g., to compile a national GHG inventory), and allows policymakers to compare emissions reduction opportunities across sectors and gases.

- CO_2, by definition, has a GWP of 1 regardless of the time period used, because it is the gas being used as the reference. CO_2 remains in the climate system for a very long time: CO_2 emissions cause increases in atmospheric concentrations of CO_2 that will last thousands of years.

- Methane (CH_4) is estimated to have a GWP of 28–36 over 100 years CH_4 emitted today lasts about a decade on average, which is much less time than CO_2. But CH_4 also absorbs much more energy than CO_2. The net effect of the shorter lifetime and higher energy absorption is reflected in the GWP. The CH_4 GWP also accounts for some indirect effects, such as the fact that CH_4 is a precursor to ozone, and ozone is itself a GHG.

- Nitrous Oxide (N_2O) has a GWP 265–298 times that of CO_2 for a 100-year timescale. N_2O emitted today remains in the atmosphere for more than 100 years, on average.

- Chlorofluorocarbons (CFCs), hydrofluorocarbons (HFCs), hydrochlorofluorocarbons (HCFCs), perfluorocarbons (PFCs), and sulfur hexafluoride (SF_6) are sometimes called high-GWP gases because, for a given amount of mass, they trap substantially more heat than CO_2. (The GWPs for these gases can be in the thousands or tens of thousands.)

Greenhouse gas	Pre-industrial concentrations*	2008 concentrations	Human source	GWP 100 years
Carbon dioxide (CO_2)	278 ppm	365 ppm	Fossil fuel combustion, land use changes, cement production	1
Methane (CH_4)	700 ppb	1745 ppb	Fossil fuels; rice paddies; waste dumps; livestock	25
Nitrous oxide (N_2O)	270 ppb	314 ppb	Fertiliser; industrial processes; fossil fuel combustion	298
Hydrofluorocarbons (e.g. HFC-23)	0	14 ppt	Liquid coolants	14,800**
Perfluorocarbons (e.g. CF_4)	0	80 ppt	Refrigerant; electronics industry and aluminium industry	6,500
Sulphur hexafluoride (SF_6)	0	4.2 ppt	Insulator in electronics and magnesium industry	22,800

* ppm, parts per million by volume; ppb, parts per billion by volume; ppt, parts per trillion by volume.

** This figure was changed in 2007 from 11,700 to 14,800.[25]

Water Vapor

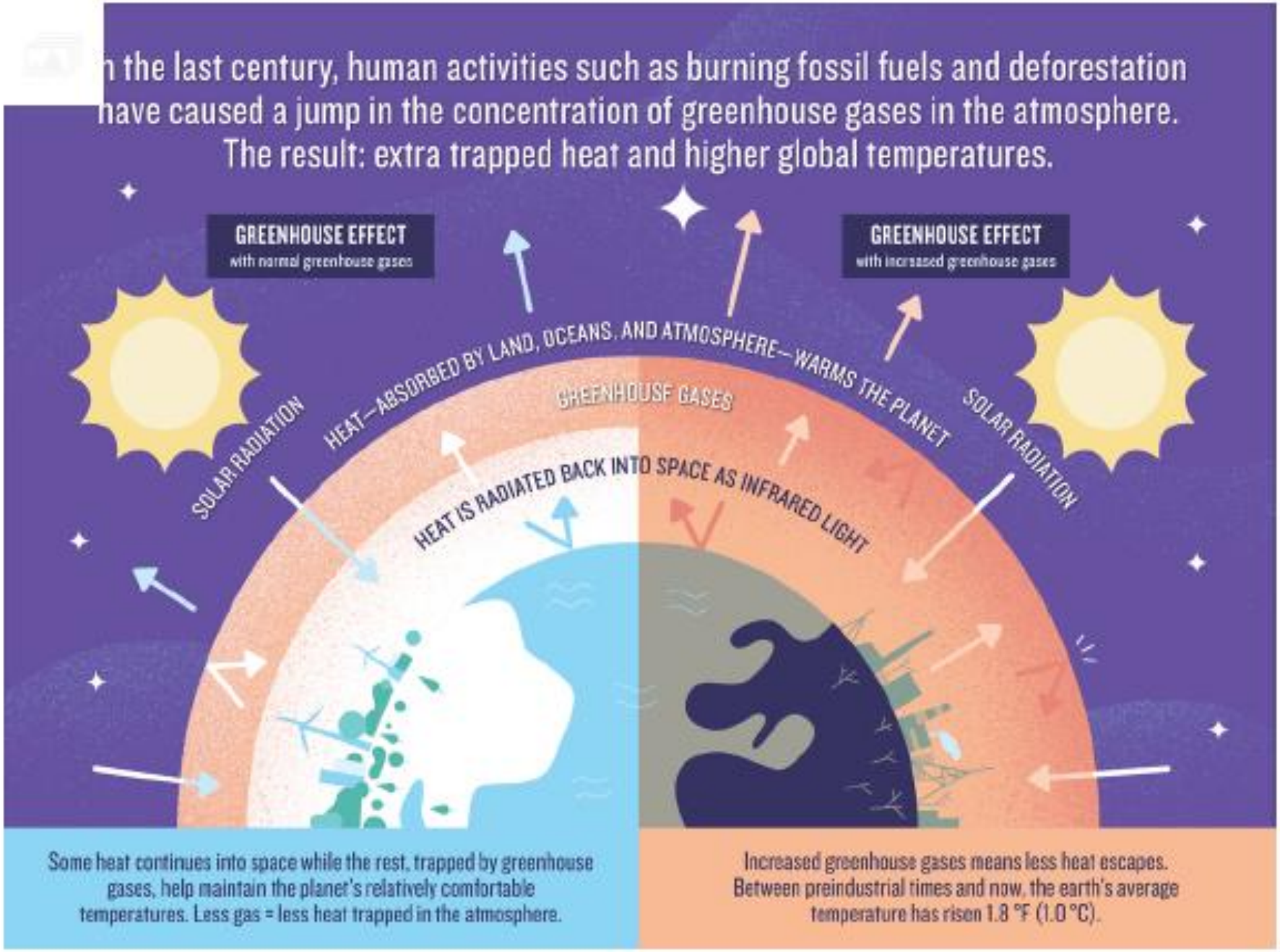
- The most abundant greenhouse gas overall, water vapor differs from other greenhouse gases in that changes in its atmospheric concentrations are linked not to human activities directly, but rather to the warming that *results* from the other greenhouse gases we emit.

- Warmer air holds more water.

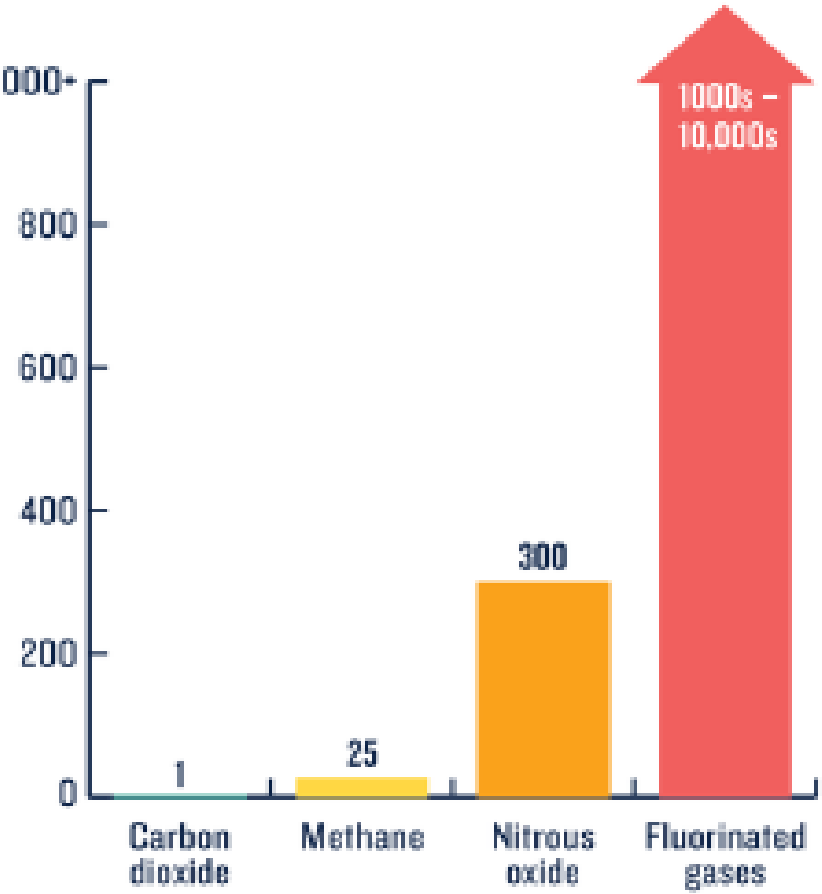
- And since water vapor is a greenhouse gas, more water absorbs more heat, inducing even greater warming and perpetuating a positive feedback loop. (It's worth noting, however, that the net impact of this feedback loop is still uncertain, as increased water vapor also increases cloud cover that reflects the sun's energy away from the earth.)

Where Do Greenhouse Gases Come From?

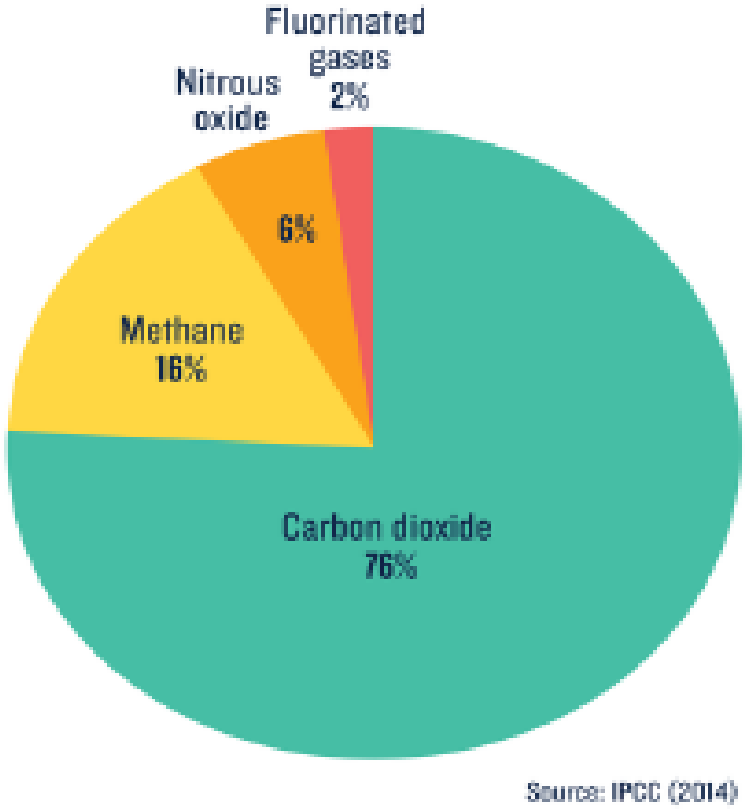
- Population size
- economic activity
- Lifestyle
- energy use
- land use patterns
- Technology
- climate policy


n the last century, human activities such as burning fossil fuels and deforestation have caused a jump in the concentration of greenhouse gases in the atmosphere. The result: extra trapped heat and higher global temperatures.
GREENHOUSE EFFECT
with normal greenhouse gases
GREENHOUSE EFFECT
with increased greenhouse gases
SOLAR RADIATION
HEAT—ABSORBED BY LAND, OCEANS, AND ATMOSPHERE—WARMS THE PLANET
GREENHOUSE GASES
SOLAR RADIATION
HEAT IS RADIATED BACK INTO SPACE AS INFRARED LIGHT
Some heat continues into space while the rest, trapped by greenhouse gases, help maintain the planet's relatively comfortable temperatures. Less gas = less heat trapped in the atmosphere.
Increased greenhouse gases means less heat escapes. Between preindustrial times and now, the earth's average temperature has risen 1.8 °F (1.0 °C).

HOW GREENHOUSE GASES WARM OUR PLANET



ne global warming potential (GWP) of human-generated greenhouse gases is a measure of how much heat each gas traps in the atmosphere, relative to carbon dioxide.



How much each human-caused greenhouse gas contributes to total emissions around the globe.

Global Warming

What is global warming?

- Over the past 50 years, the average global temperature has increased at the fastest rate in recorded history.

- One of the 16 hottest years in NASA's 134-year record have occurred since 2000.

- Climate change deniers have argued that there has been a "pause" or a "slowdown" in rising global temperatures.

What causes global warming?

- Carbon dioxide (CO2) and other air pollutants and greenhouse gases
- They collect in the atmosphere and absorb sunlight and solar radiation that have bounced off the earth's surface.
- This can cause earth's rising temperatures with hotter heat waves, more frequent droughts, heavier rainfall, and more powerful hurricanes.
- Melting glaciers, early snowmelt, and severe droughts will cause more dramatic water shortages and increase the risk of wildfires.
- Rising sea levels will lead to coastal flooding Forests, farms, and cities will face troublesome new pests, heat waves, heavy downpours, and increased flooding.
- All those factors will damage or destroy agriculture and fisheries.
- Disruption of habitats such as coral reefs and Alpine meadows could drive many plant and animal species to extinction.
- Allergies, asthma, and infectious disease outbreaks will become more common due to increased growth of pollen-producing ragweed, higher levels of air pollution, and the spread of conditions favorable to pathogens and mosquitoes.

How can I reduce Global Warming

- Reduce your own carbon footprint by following a few easy steps.
- Make conserving energy a part of your daily routine and your decisions as a consumer.
- When you shop for new appliances like refrigerators, washers, and dryers, look for products with the government's Energy Star label; they meet a higher standard for energy efficiency than the minimum federal requirements.
- When you buy a car, look for one with the highest gas mileage and lowest emissions.
- You can also reduce your emissions by taking public transportation or carpooling when possible.

Reduce Global Warming

- **Reduce, Reuse, Recycle**
- **Use Less Heat and Air Conditioning**
- **Replace Your Light Bulbs**
- **Drive Less and Drive Smart**
- **Buy Energy-Efficient Products**
- **Use Less Hot Water**
- **Use the "Off" Switch**
- **Plant a Tree**
- **Get a Report Card from Your Utility Company**
- **Encourage Others to Conserve**

The global climate’s heat-controlling systems and subsystems

- **Oceans with their currents, different water temperatures, and descending and ascending layers hold absorbed carbon or heat.**

Initially, the oceans absorb carbon and help us.

But when too much carbon is absorbed, the oceans begin the process of emitting carbon back into the atmosphere.

That will raise temperatures.

- **Forests can either eat or release carbon** based on the temperature and other conditions.
- When trees die, their stored carbon is released back into the atmosphere.
- Trees normally take carbon out of the atmosphere.
- If certain conditions exist or it gets too warm trees will take less carbon out of the air.

- **Soils can also eat or release carbon** depending upon their condition under heat variables.

- This is due to carbon deposits from plant life.

- **The carbon-eating and oxygen-producing plankton in the oceans**.
- If the oceans absorb too much carbon from global warming, they become acidic—specifically carbonic acid.
- This acidity will eventually kill some or all of the carbon-eating and oxygen-producing plankton.
- If we kill off this necessary plankton, we will find ourselves with insufficient oxygen in a world no one will be able to endure.

- The **carbon and methane-releasing volcanoes**.
- Sustained large-scale volcanic activity can drastically affect the environment.
- If the volcano is large enough, such as with a super volcano, the eruption could actually cool the planet and create two or three years of nuclear winter.
- Such a development creates its own extinction-level destruction in the form of severe negative impacts on agriculture and other living systems.

- Today's global warming emergency is not a natural disaster. It is a human-made disaster.

- A small increase in average global temperature will eventually create catastrophic changes in the world.

- As global warming and climate destabilization continue, our local and national weather, as well as our global climate, will become much more unpredictable. Storms, droughts, floods, seasonal disruptions, sea-level rise, and wildfires are going to become more severe, frequent, and occur at larger scales.

- As the global climate continues to destabilize, most people will not realize their lives are also destabilizing after it is too late.

- We may have already reached irreversible climate destabilization. It will go on much longer than our human life spans. It can last for centuries or thousands of years.

- A continually destabilizing climate due tp escalating global warming will be the greatest disruptor of normal life that we have ever known—exceeding even our greatest world wars. Get up there me to help you